Math 205A - Fall 2014
Homework #4
Solutions

Problem 1: Let $f \in L^p$ and $\mu(t) = m\{x : |f(x)| > t\}$ the distribution function of $|f|$. Show that:
(i) $\mu(t) \leq \frac{1}{t^p}||f||^p_{L^p(\Omega)}$.
(ii) $||f||^p_{L^p(\Omega)} = p\int_0^\infty t^{p-1}\mu(t)dt$.
(iii) For any increasing differentiable function ϕ with $\phi(0) = 0$: $\int_\Omega \phi(|f(x)|)dx = \int_0^\infty \phi'(\lambda)\mu(\lambda)d\lambda$.

(i) $||f||^p_{L^p(\Omega)} \geq \int\limits_{\{|f(x)|>t\}} |f|^p \geq \int\limits_{\{|f(x)|>t\}} t^p = t^p \int\limits_{\{|f(x)|>t\}} |f|^p = t^p\mu(t)$.

(ii) Follows from (iii).

(iii) We'll prove the result by applying problem 4 (Fubini) to $\int\limits_\Omega \int\limits_0^\infty \phi'(t)1_{[0,|f(x)|]}(t)dtdx$:

$$\int\limits_\Omega \left(\int\limits_0^\infty \phi'(t)1_{[0,|f(x)|]}(t)dt \right) dx = \int\limits_\Omega \left(\int\limits_0^{|f(x)|} \phi'(t)dt \right) dx = \int\limits_\Omega \phi(|f(x)|)dx,$$

and for the left hand side

$$\int\limits_0^\infty \left(\int\limits_\Omega \phi'(t)1_{[0,|f(x)|]}(t)dx \right) dt = \int\limits_0^\infty \phi'(t) \left(\int\limits_\Omega 1_{[0,|f(x)|]}(t)dx \right) dt = \int\limits_0^\infty \phi'(t)\mu(t)dt.$$

□

Problem 2: Show that for $f \in ||f||_{L^p}$, $1 \leq p < \infty$ and $\phi \in L^1$, $\phi \geq 0$, $\int \phi = 1$, if $\phi_t(x) = t^{-1}\phi(x/t)$, then $\lim\limits_{t\to 0} ||\phi_t \star f - f||_p = 0$.

The crucial step is to prove that $||f * g||_p \leq ||f||_p||g||_1$:

$$\left| \int f(x-y)g(y)\,dy \right| \leq \int |f(x-y)|\,|g(y)|^{\frac{1}{p}}\,|g(y)|^{\frac{p-1}{p}}\,dy \leq ||\,|f(x-\cdot)|\,|g|^{\frac{1}{p}}||_p\,||\,|g|^{\frac{p-1}{p}}||_{\frac{p}{p-1}} = ||\,|f(x-\cdot)|\,|g|^{\frac{1}{p}}||_p\,||g||_1^{\frac{p-1}{p}},$$

therefore

$$||f * g||_p^p \leq \int \left(||\,|f(x-\cdot)|\,|g|^{\frac{1}{p}}||_p\,||g||_1^{\frac{p-1}{p}} \right)^p dx = ||g||_1^{p-1} \int \left(\int |f(x-y)|^p |g(y)|\,dy \right) dx \tag{1}$$

$$= ||g||_1^{p-1} \iint |f(x-y)|^p |g(y)|\,dx\,dy = ||g||_1^{p-1}||f||_p^p||g||_1 = ||g||_1^p||f||_p^p. \tag{2}$$

Now if the result is true for step functions, then since they are dense in L^p, there exists a sequence f_n of such functions with $f_n \xrightarrow{L^p} f$. But then

$$\begin{aligned} ||\phi_t \star f - f||_p \leq & \; ||\phi_t \star f - \phi_t \star f_n||_p + ||\phi_t \star f_n - f_n||_p + ||f_n - f||_p \\ \leq & \; ||\phi_t||_1||f_n - f||_p + ||\phi_t \star f_n - f_n||_p + ||f_n - f||_p \\ \leq & \; ||f_n - f||_p + ||\phi_t \star f_n - f_n||_p + ||f_n - f||_p. \end{aligned}$$

Given ϵ, choose n such that $||f_n - f||_p < \epsilon/2$. In the limit, the 2nd term vanishes, therefore $\lim\limits_{t\to 0} ||\phi_t \star f - f||_p \leq \epsilon$. So the final step is to prove the claim for step functions. By triangle inequality, it's enough to consider $f = 1_{[a,b]}$.

$$||\phi_t \star f - f||_p^p = \int \left(\frac{1}{t} \int \phi(\frac{x-y}{t}) f(y) dy - f(x) \right)^p dx = \int \left(\int \phi(z) f(x-tz) dz - f(x) \right)^p dx$$
$$= \int \left(\int \phi(z)(f(x-tz) - f(x)) dz \right)^p dx$$
$$= \iint \phi(z) |f(x-tz) - f(x)|^p dz\, dx. \qquad \longleftarrow \text{ the same proof as for (2) above}$$

Now given ϵ, pick R s.t. $\int_{|z| \geq R} \phi(z) \leq \frac{\epsilon}{2(b-a)}$. Note that $|f(x-tz) - f(x)|^p = |f(x-tz) - f(x)| \in \{0, 1\}$, and $\int |f(x-tz) - f(x)|^p \leq 2(b-a)$. Using these, for $t < \frac{b-a}{R}$:

$$\iint \phi(z) |f(x-tz) - f(x)|^p dz\, dx \leq \epsilon + \int\limits_{[-R,0]} \int \phi(z) |f(x-tz) - f(x)| dx\, dz + \int\limits_{[0,R]} \int \phi(z) |f(x-tz) - f(x)| dx\, dz.$$

$$\text{But} \quad \int\limits_0^R \int \phi(z) |f(x-tz) - f(x)| dx\, dz = \int\limits_0^R \phi(z) \left(\int\limits_a^{a+tz} dx + \int\limits_b^{b+tz} dx \right) dz \leq 2Rt \int\limits_0^R \phi(z) dz \leq 2Rt \longrightarrow 0 \quad \text{as } t \to 0,$$

and similarly the other integral. Hence, $\lim ||\phi_t \star f - f||_p^p \leq \epsilon$. This finishes the proof.
Note: The first inequality is a special case of Young's inequality: $||f * g||_r \leq ||f||_p ||g||_q$ for $\frac{1}{p} + \frac{1}{q} = \frac{1}{r} + 1$. □

Problem 3: Let $\mu(X) < \infty$, $f \in L^\infty(X, d\mu)$, and $\alpha_n = \int_X |f|^n d\mu$ for $n \in \mathbb{N}$. Prove that $\lim_{n\to\infty} \frac{\alpha_{n+1}}{\alpha_n} = ||f||_\infty$.

Clearly $\alpha_{n+1} \leq ||f||_\infty \alpha_n$, so $\lim \frac{\alpha_{n+1}}{\alpha_n} \leq ||f||_\infty$. On the other hand, using Hölder:

$$\alpha_n = \int |f|^n d\mu \leq || |f|^n ||_{\frac{n+1}{n}} \cdot ||1_X||_{n+1} = \left(\int |f|^{n+1} d\mu \right)^{\frac{n}{n+1}} \mu(X)^{\frac{1}{n+1}} = \alpha_{n+1}^{\frac{n}{n+1}} \mu(X)^{\frac{1}{n+1}} \implies \frac{\alpha_{n+1}}{\alpha_n} \geq \left(\frac{\alpha_n}{\mu(X)} \right)^{\frac{1}{n}}.$$

Now given $0 < \epsilon < ||f||_\infty$, using problem 1: $\alpha_n^n \geq (||f||_\infty - \epsilon)^n \mu(\{x : |f| \geq ||f||_\infty - \epsilon\})$, so taking the limit,

$$\lim_{n\to\infty} \frac{\alpha_{n+1}}{\alpha_n} \geq \lim_{n\to\infty} (||f||_\infty - \epsilon) \left(\frac{\mu(\{x : |f| \geq ||f||_\infty - \epsilon\})}{\mu(X)} \right)^{\frac{1}{n}} = ||f||_\infty - \epsilon,$$

because $\frac{\mu(\{x : |f| \geq ||f||_\infty - \epsilon\})}{\mu(X)} > 0$ and does not depend on n. Since this is true for every ϵ, sending $\epsilon \to 0$ we get that $\lim \frac{\alpha_{n+1}}{\alpha_n} \geq ||f||_\infty$ as well. □

Problem 4: Let $f \geq 0$ be a measurable function on $X \times Y$. Show that then:
(i) for almost all x the function $f(x, y)$ is ν-measurable (as a function of y with x being a parameter) and for almost all y the function $f(x, y)$ is μ-measurable (as a function of x with y being a parameter);
(ii) $\int f(x, y) d\mu(x)$ is measurable and $\int f(x, y) d\nu(y)$ is measurable;
(iii) $\int_{X\times Y} f d(\mu \times \nu) = \int_X \left[\int_Y f d\nu \right] d\mu = \int_Y \left[\int_X f d\mu \right] d\nu$;
(iv) Show that the assumption that f is non-negative may not be removed: take $X = Y = \mathbb{N}$, let $\mu = \nu$ be the counting measure and let

$$f(n, m) = \begin{cases} 1, & n = m \\ -1, & n = m + 1 \\ 0, & \text{otherwise} \end{cases}$$

Show that $\int |f| d(\mu \times \nu) = \infty$ while $\int (\int f d\mu) d\nu$ and $\int (\int f d\nu) d\mu$ both exist and are not equal.

(i), **(ii)** and **(iii)** follow essentially the same lines as chapter 5 of the notes, so refer to them for the proofs.
(iv) Note first that $|f| = 1$ on (m, m) and $(m+1, m)$, and there are infinitely many such points, so $\int |f| d(\mu \times \nu) = \infty$.

On the other hand, $\int f(n,m)d\mu(n) = f(m,m) + f(m+1,m) = 0$, so $\int(\int f d\mu)d\nu = 0$, whereas $\int f(n,m)d\nu(m) = f(n,n) + f(n-1,n) = 0$ for $n \geq 2$, and $\int f(1,m)d\nu(m) = f(1,1) = 1$, so $\int(\int f d\nu)d\mu = 1$. □

Problem 5: Let

$$\phi_0(t) = \begin{cases} 1, & x \in [0,1) \\ -1, & x \in [1,2), \end{cases}$$

extend it periodically to all of $\mathbb{R}$, and define $\phi_n(t) = \phi_0(2^n t)$, $n \in \mathbb{N}$. Assume that $\sum |c_n|^2 < \infty$ and show that the series $\sum c_n \phi_n(t)$ converges for almost every t.

Because of the 1-periodicity of ϕ_n, $n \geq 1$, it's enough to prove the statement for a.e. $t \in [0,1]$. Let $s_n(t) = \sum_1^n c_j \phi_j(t)$.

We want to show that $\{s_n\}$ is almost everywhere Cauchy (hence a.e. convergent). So let E be the 'bad' set where this doesn't happen:

$$E = \bigcup_m \bigcap_N \underbrace{\bigcup_{n \geq N+1} \{t : |s_n(t) - s_N(t)| \geq \frac{1}{m}\}}_{E_{N,1/m}}$$

If we show that $m(\bigcap_N E_{N,\epsilon}) = 0$ for all $\epsilon > 0$, then $m(E) \leq \sum m(\bigcap_N E_{N,1/m}) = 0$. Now $m(\bigcap_N E_{N,\epsilon}) \leq m(E_{N,\epsilon})$ for all N, so $m(\bigcap_N E_{N,\epsilon}) \leq \lim_{N\to\infty} m(E_{N,\epsilon})$. On the other hand,

$$\begin{aligned}
m(E_{N,\epsilon}) = &\quad m(\{t : \sup_{n \geq N+1} |s_n(t) - s_N(t)| \geq \epsilon\}) \\
= &\quad m(\{t : \sup_{n \geq N+1} \max_{N+1 \leq k \leq n} |s_k(t) - s_N(t)| \geq \epsilon\}) \\
= &\quad m(\bigcup_{n \geq N+1} \{t : \max_{N+1 \leq k \leq n} |s_k(t) - s_N(t)| \geq \epsilon\}) \\
= &\quad \lim_{n\to\infty} m(\{t : \max_{N+1 \leq k \leq n} |s_k(t) - s_N(t)| \geq \epsilon\}) \qquad \leftarrow \text{because the sets are increasing}
\end{aligned}$$

Let $T(t) \geq N+1$ be the first index k for which $|s_k(t) - s_N(t)| \geq \epsilon$. Then

$$\begin{aligned}
m(\{t : \max_{N+1 \leq k \leq n} |s_k(t) - s_N(t)| \geq \epsilon\}) = &\quad m(\{t : N+1 \leq T(t) \leq n\}) \\
= &\quad \int_0^1 1_{\{t : N+1 \leq T(t) \leq n\}} dt \\
\leq &\quad \tfrac{1}{\epsilon^2} \int_0^1 |s_{T(t)} - s_N|^2 1_{\{N+1 \leq T(t) \leq n\}} dt \qquad \longleftarrow |s_{T(t)} - s_N| \geq \epsilon \\
= &\quad \tfrac{1}{\epsilon^2} \sum_{k=N+1}^n \int_0^1 |s_k - s_N|^2 1_{\{T(t)=k\}} dt \\
= &\quad \tfrac{1}{\epsilon^2} \sum_{k=N+1}^n \int_0^1 |\sum_{i=N+1}^k c_i \phi_i(t)|^2 1_{\{T(t)=k\}} dt \\
= &\quad \tfrac{1}{\epsilon^2} \sum_{k=N+1}^n \int_0^1 \sum_{i,j=N+1}^k c_i \bar{c}_j \phi_i(t) \bar{\phi}_j(t) 1_{\{T(t)=k\}} dt \\
= &\quad \tfrac{1}{\epsilon^2} \sum_{k=N+1}^n \{\int_0^1 \sum_{i=N+1}^k |c_i|^2 \underbrace{|\phi_i(t)|^2}_{=1} 1_{\{T(t)=k\}} + \underbrace{\int_0^1 \sum_{i \neq j} c_i \bar{c}_j \phi_i(t) \bar{\phi}_j(t) 1_{\{T(t)=k\}}\}}_{=0 \text{ (easy)}} \\
= &\quad \tfrac{1}{\epsilon^2} \int_0^1 \sum_{k=N+1}^n \sum_{i=N+1}^k |c_i|^2 1_{\{T(t)=k\}} dt \\
= &\quad \tfrac{1}{\epsilon^2} \int_0^1 \sum_{i=N+1}^n |c_i|^2 1_{\{N+1 \leq T(t) \leq n\}} dt \leq \tfrac{1}{\epsilon^2} \sum_{i=N+1}^n |c_i|^2.
\end{aligned}$$

Hence,

$$\begin{aligned} m(\textstyle\bigcap_N E_{N,\epsilon}) \leq & \lim_{N\to\infty} m(E_{N,\epsilon}) = \lim_{N\to\infty}\lim_{n\to\infty} m(\{\max_{N+1\leq k\leq n} |s_k(t)-s_N(t)| \geq \epsilon\}) \\ \leq & \lim_{N\to\infty}\lim_{n\to\infty} \tfrac{1}{\epsilon^2}\sum_{i=N+1}^{n} |c_i|^2 = \tfrac{1}{\epsilon^2}\lim_{N\to\infty}\sum_{i=N+1}^{\infty} |c_i|^2 \\ = & \ 0 \qquad\qquad \longleftarrow \text{because } \textstyle\sum |c_i|^2 < \infty. \end{aligned}$$

Therefore $m(E) = 0$ and $\{s_n(t)\}$ converges for almost every t. □